Rupesh G

Tamil Nadu, India

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Education

Bachelor of Technology in Mechanical Engineering — Aug 2023 - Present
National Institute of Technology, Tiruchirappalli (NIT Trichy) — *Tamil Nadu, India*

- **CGPA: 8.75/10** (Through Semester VI)
- **Minor:** Computer Science Engineering

Class XII (Higher Secondary Education) — May 2022 - July 2023
The Pupil Saveetha Eco School, Chennai — *Tamil Nadu, India*

- **Percentage: 96.60%**

Relevant Coursework

Curriculum: Mechanics of Machines, Engineering Mechanics, Strength of Materials, Fluid Mechanics, Design of Machine Elements, Vehicle Dynamics, Computer-Aided Design, Mechatronics, MEMS Design and Fabrication, Mathematical Methods for Mechanical Engineers, Matrices and Calculus, Complex Analysis and Differential Equations, Fourier Transforms and Numerical Techniques, Applied Electrical and Electronics Engineering, Artificial Intelligence, Data Structures and Algorithms.

Other Certifications: MATLAB Onramp, Simulink Onramp, Introduction to IoT, Introduction to Machine Learning, Deep Learning Onramp.

Internships

ROS 2-Based Robotic Motion Planning and Simulation — May 2026 – July 2026
Engineering Design and Research Center (EDRC), Larsen & Toubro (L&T) — *Tamil Nadu, India*

- Developed **ROS 2-based robotic simulation workflows** integrating **MoveIt and RViz** for motion planning, trajectory visualisation, and system validation.
- Built and configured **URDF robot models** to represent robot kinematics and support simulation, motion planning, and trajectory execution.
- Configured and evaluated **robot motion-planning pipelines** in simulated environments to assess trajectory feasibility and robot behaviour.
- Collaborated with robotics engineers on **software integration, simulation testing, and technical documentation** within an industrial R&D environment focused on robotics and automation.

Phase-Aware Multi-UAV Coordination for Urban Air Mobility — May 2025 – June 2025
Dr. Sunetra Sarkar, Biomimetics and Dynamics Laboratory, IIT Madras — *Tamil Nadu, India*

- Modelled and simulated a **16-UAV urban air mobility system** in **MATLAB/Simulink**, implementing coordinated flight across 4 routes.
- Implemented a **leader–follower architecture** with 4 leaders and 12 followers, developing control strategies for **take-off, cruise, and landing**.
- Performed **controller tuning and time-domain response analysis** to evaluate trajectory tracking, bank-angle behaviour, and system stability under simulated flight conditions.

Projects

Mobile Manipulator — Jul 2026 – Present

- Designed and 3D-printed a **mobile robotic platform** integrated with a **6-DOF robotic arm**, covering mechanical design and physical assembly.
- Implemented **open-loop actuator control** and completed initial hardware-level testing of the robotic system.
- Created the robot's **URDF model** and completed visualization in **RViz**; currently integrating the model with **ROS 2** for simulation and control.
- Developing an **OpenCV-based object detection pipeline** to support visual perception for robotic manipulation.

Neck Rehabilitation Robot Jan 2026 – Present

- Developed a **6-DOF Stewart platform mechanism** for controlled robotic cervical rehabilitation.
- Designed the **metal mechanism** in **Fusion 360** to provide controlled **pitch, roll, and yaw** movements.
- Developed the control system in **Simulink** with headband-mounted sensors for **real-time motion feedback**.
- Currently working on **electronics integration**, including actuator control, sensor interfacing, and feedback implementation.

Balancing Builder Bot Sep 2024 – Feb 2025

- Designed a **two-wheeled self-balancing pick-and-place robot** with a gripper using SolidWorks.
- Modelled robot dynamics using the **inverted pendulum principle**, deriving a state-space representation and implementing an **LQR controller** in MATLAB.
- Achieved over **90% balancing efficiency in CoppeliaSim simulations** through controller tuning and evaluation of robot stability.

Robot Pathfinding and Multi-Agent Path Planning Sep 2026 – Oct 2026

- Implemented **A* pathfinding** with 4 heuristic strategies—Manhattan, Euclidean, Chebyshev, and zero heuristic—to evaluate path cost and search efficiency in obstacle-filled grids.
- Implemented and compared **4 multi-agent path-planning algorithms**: Independent A*, Prioritized Planning, Conflict-Based Search (CBS), and a proposed **Hybrid Prioritized Planning + CBS (H-PCBS)** approach.
- Developed a **benchmarking framework** to evaluate runtime, success rate, sum of costs (SOC), makespan, and collision conflicts across varying grid sizes, obstacle densities, and robot counts.

Skill Summary

- **Programming:** Python, C, C++, MySQL, HTML, CSS
- **Robotics & Simulation:** ROS 2, MoveIt, Gazebo, RViz, URDF, CoppeliaSim, OpenCV, Machine Learning
- **CAD, Design & Control Systems:** SolidWorks, Fusion 360, AutoCAD, ANSYS, MATLAB, Simulink, PID Control, LQR, State-Space Modelling
- **Other Software:** Microsoft Word, Excel, PowerPoint, Visual Studio, Photoshop

Executive Positions

President & Robotics Engineer, Spider R&D Aug 2024 – Present
Official Research and Development Club *NIT Trichy, India*

- Lead the club's **hardware and software divisions**, overseeing team operations, project planning, resource allocation, and technical initiatives.
- Drive **end-to-end robotics projects** spanning mechanical design, embedded systems, control engineering, and simulation, from concept to integration.
- Mentor junior members in **robotics, CAD, and control systems** through technical workshops and hands-on training.

Head, Networking & Speakers Forum, Synergy Dec 2024 – Present
Mechanical Engineering Symposium *NIT Trichy, India*

- Sent **20+ speaker pitches**, achieving a **15% conversion rate** and engaging senior professionals from leading automotive companies.
- Organize and moderate **3 technical lectures** for **300+ students each**, managing speaker coordination and event execution.

Design Manager, Nittfest Aug 2024 – Jul 2025
Interdepartmental Cultural Fest *NIT Trichy, India*

- Led a team of **12 coordinators** responsible for graphic design and visual communication for the fest.
- Created **20+ event thumbnails and 10+ social media posts** using **Photoshop**, supporting promotional campaigns and fest branding.

Public Relations Manager, Festember Dec 2024 – Present
Annual Inter-College Cultural Fest *NIT Trichy, India*

- Manage **guest accommodation and hospitality logistics**, including hotel arrangements and guest support.
- Oversee the **helpline and PR desk**, coordinating guest communication and on-ground operations.